

DS	Session	Searching	Question Information	Level 1	Challenge 10
Question Description: An array A contains integers with the following constraints: 'A' contains elements in sorted order. Integer i occurs $i \cdot \text{floor}(\sqrt{i}) + \text{ceil}(i/2)$ times in the array. All elements are greater than or equal to 1. Fathima has given Q queries of type: L R: Find the number of distinct values present in subarray A[L...R]. Note: 1-based indexing is followed. Constraints: $1 \leq Q \leq 10^5$ $1 \leq L \leq R \leq 10^{13}$ Input Format: The first line contains an integer Q denoting the number of queries. Next Q lines contains two space-separated integers L, R, denoting the query. Output Format: Print the output in separate lines contains required number of distinct values. Example Sample Input: 2 1 3 1 6 Sample Output: 2					

Explanation

First few elements of the array A are 1, 1, 2, 2, 2, 2, 3, 3, 3, 3, 3, ...

- For Query 1:-
 - Number of distinct elements in subarray $A[1...3]$ is 2.
- For Query 2:-
 - Number of distinct elements in subarray $A[1...6]$ is 3.

Logical Test Cases

Test Case 1

INPUT (STDIN)

2
3 3
2 6

EXPECTED OUTPUT

1
3

Test Case 2

INPUT (STDIN)

3
3 3
2 6
5 4

EXPECTED OUTPUT

1
3
1

Mandatory Test Cases

Test Case 1

KEYWORD

while(l<ans1)

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

6

Test Case 2

TOKEN COUNT

400

Test Case 3

NLOC

56

Searching10.c +

```
1 #include <stdio.h>
2
3 long long get_sum(long long n) {
4     long long sum_i = n * (n + 1) / 2;
5     long long sum_ceil;
6     if (n % 2 == 0) {
7         long long k = n / 2;
8         sum_ceil = k * (k + 1);
9     } else {
10         long long k = (n + 1) / 2;
11         sum_ceil = k * k;
12     }
13     return sum_i + sum_ceil;
14}
15
16 long long get_val(long long pos) {
17     long long low = 1, high = 50000000;
18     long long ans = high;
19     while (low <= high) {
20         long long mid = low + (high - low) / 2;
21         if (get_sum(mid) >= pos) {
22             ans = mid;
23             high = mid - 1;
24         } else {
```

```
26     }
27 }
28 return ans;
29}
30
31int main() {
32     int q;
33     if (scanf("%d", &q) != 1) return 0;
34     while (q--) {
35         long long l, r;
36         scanf("%lld %lld", &l, &r);
37         if (l > r) {
```

```
38     long long tmp = 1;
39     l = r;
40     r = tmp;
41 }
42 long long v1 = get_val(l);
43 long long v2 = get_val(r);
44 printf("%lld\n", v2 - v1 + 1);
45 }
46 return 0;
47}
48
```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

2

3 3

2 6

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Searching10.exe

1

3

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

3

3 3

2 6

5 4

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Searching10.exe

1

3

1

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>